

Name: _____ Date: _____

Period: _____

1. For each transformation rule below, identify the type of transformation (vertical translation, horizontal translation, vertical dilation, or horizontal dilation/reflection) and describe its effect on the graph of f .

(a) $g(x) = f(x) - 5$

Type: _____ Effect: _____

(b) $g(x) = f(x - 3)$

Type: _____ Effect: _____

(c) $g(x) = \frac{1}{3}f(x)$

Type: _____ Effect: _____

(d) $g(x) = f(-x)$

Type: _____ Effect: _____

2. The table below gives values of f .

x	-2	-1	0	1	2
$f(x)$	8	2	0	2	8

Complete a table of values for each transformed function.

(a) $g(x) = f(x) + 4$

x	-2	-1	0	1	2
$g(x)$	_____	_____	_____	_____	_____

(b) $h(x) = -f(x)$

x	-2	-1	0	1	2
$h(x)$	_____	_____	_____	_____	_____

3. Let f have domain $[1, 5]$ and range $[-3, 6]$. For each transformed function, state the domain and range.

(a) $g(x) = f(x) + 2$ Domain: _____ Range: _____

(b) $g(x) = f(x + 4)$ Domain: _____ Range: _____

(c) $g(x) = 3f(x)$ Domain: _____ Range: _____

(d) $g(x) = f(x - 1) - 2$ Domain: _____ Range: _____

4. Describe all transformations applied to f to obtain $g(x) = -3f(x + 2) + 1$. List them in order of application.

5. **(AP-Style Multiple Choice)** Let f be a function with domain $[-4, 4]$ and range $[0, 8]$. Which of the following gives the range of $g(x) = -2f(x) + 1$?

- (A) $[-15, 1]$
- (B) $[-16, 0]$
- (C) $[1, 17]$
- (D) $[-15, -1]$

Answer: _____ Show your reasoning:

Reflection

What is the trickiest part of horizontal transformations? Why does $f(x + h)$ shift *opposite* to the sign of h ?
